



[30 x 1pt]

Question #1: Multiple choices.

- The average-case time complexity of sequential search is
 A) n B) 1 C) $(n+1)/2$ D) $\log n + 1$
- If the size of an array is 512, then the total number of comparisons done by binary search and sequential search are respectively.
 A) 10, 512 B) 9, 512 C) 512, 10 D) 512, 9
- Given the below sorted array:
 $arr[] = \{1, 14, 16, 20, 24, 26, 30, 40, 45, 49, 50, 55, 70, 97, 102, 107, 110\}$
 To determine if $x = 25$ exists in the above array using the binary search algorithm, the numbers compared in order with x are:
 A) 45, 24, 30 B) 45, 24, 30, 26 C) 45, 20, 26, 24 D) 45, 20, 26
- What is the number of terms computed to find the $fib(5)$ using the recursive solution (algorithm) of the Fibonacci sequence?
 A) 7 B) 15 C) 25 D) 9

5. Find the time complexity of this function .

```
void fun(int n) {
    if (n==1) {
        cout<<"*";
        cout<<"*";
    }
    else {
        cout<<"*";
        fun(n-1);
    }
}
```

- A) $T(n) = T(n-1) + 2$ $T(1) = 1$;
 B) $T(n) = T(n-1) + 1$ $T(1) = 2$;
 C) $T(n) = 2T(n-1) + 2$ $T(1) = 0$;
 D) $T(n) = T(n) + 3$ $T(1) = 0$;

6. Find the time complexity of print function .

```
void fun(int a[], int m) {
    for (i=0; i<m*2; i++)
        cout<<a[i];
}

int print (int a[], n) {
    fun(int a[], n/2);
    for (i=1; i<n*3; i++)
        cout<a[i];
    fun(int a[], n/2);
}
```

- A) $3n - 1$ B) $4n - 1$
 C) $5n - 1$ D) $7n - 1$

$$m = \frac{n}{2}$$

$$2m + 3n - 1 + 2m$$

$$n + 3n - 1 + n = 5n - 1$$

- In Mergesort to merge two subarrays $U = [5, 7, 8, 10]$ and $V = [4, 9, 11, 12]$ into one array S , how many times the basic operation (comparison) in the merge algorithm will be executed?
 A) 5 B) 6 C) 7 D) 8
- Which sorting algorithm has the least worst case time complexity
 A) Exchange sort B) Merge sort C) Quick sort D) all the same

*** Answer the following 2 questions using quick sort to sort the following array $S = \{9, 16, 4, 23, 1, 8, 14, 7\}$. Choose the pivot item to be the first item,;



9. time complexity for partition function is

A) $h+m-1$

B) $n-1$

C) $n * \log n$

D) n^2

10. In the first step after dividing the array S will be _____

A) {1,4,7,8,9,16,14,23}

B) {7,4,1,8,9,23,14,16}

C) {4,1,8,7,9,16,23,14}

D) None of the above

*** Answer the following 3 questions for the given below recurrence equation:

$$t_n + 3t_{n-1} - 4t_{n-2} = 0 \quad t_0 = 0, \quad t_1 = 5$$

11. What is the characteristic equation?

A) $r^2 + 3r - 4 = 0$

B) $r^2 - 3r + 4 = 0$

C) $-4r^2 + 3r + 1 = 0$

D) $4r^2 + 3r - 1 = 0$

12. What are the roots of the characteristic equation?

A) $r_1 = 1$ and $r_2 = 4$

B) $r_1 = 1$ and $r_2 = -4$

C) $r_1 = -1$ and $r_2 = -4$

D) $r_1 = -1$ and $r_2 = 4$

13. What is the particular form of the solution of the recurrence equation?

A) $t_n = -1 - (4)^n$

B) $t_n = 1 + (4)^n$

C) $t_n = 1 - (-4)^n$

D) $t_n = -1 + (-4)^n$

14. The solution of the following recurrence is: $T(n) = T(n-1) + 6n, T(1) = 4$

A) $6 * n - 2$

B) $6 * n - 4$

C) $3 * n^2 + 3 * n - 2$

D) $3 * n^2 + 6 * n - 4$

15. The big theta of $F(n) = 8F\left(\frac{n}{3}\right) + n^2$

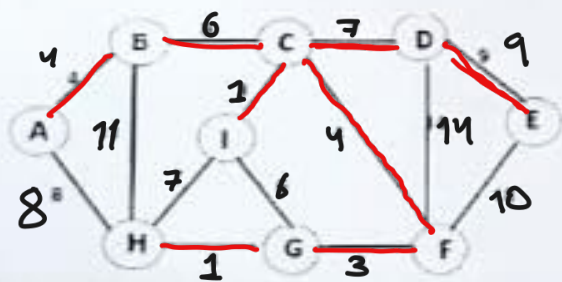
A) $n^2 * \log n$

B) n^2

C) $n^3 * \log n$

D) n^3

16. use the Kruskal's algorithm to find the cost of the minimum spanning tree for the following graph



A) 38

B) 39

C) 45

D) 35

17. Prim's algorithm is used for _____

A) Finding the Longest common subsequence.

B) Finding the shortest path between every pair of vertices in a (di)graph.

C) Finding the minimum spanning tree.

D) None of the above

18. Floyd's algorithm is a _____

A) Divide and conquer algorithm

B) Greedy algorithm

C) Dynamic programming algorithm

D) Backtracking algorithm



*** Suppose there are five jobs and the service time of these jobs are $t_1 = 18$, $t_2 = 12$, $t_3 = 10$, $t_4 = 23$, and $t_5 = 8$, answer the following 2 questions:

19. The job's schedule that minimizes the total time spent in the system is:
 A) [2, 5, 1, 3, 4] B) [2, 5, 1, 4, 3] C) [5, 3, 2, 1, 4] D) [5, 4, 3, 2, 1]

20. The total time spent in the system for the optimal schedule is:
 A) 175 B) 202 C) 204 D) None of the above

21. Based on the information given in the next table, which of the following could be a feasible sequence generated when solving the job scheduling with deadlines problem using the greedy method?

Job	Deadline	Profit
1	4	50
2	1	100
3	2	75
4	4	120
5	3	150
6	2	60
7	1	80

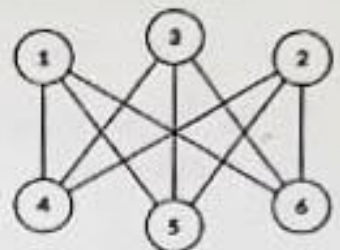
- A) [5, 4, 2, 7] B) [5, 4, 2, 3]
 C) [2, 3, 5, 4] D) None of the above

22. Which of the following encoding is an example of binary prefix-free code?

- A) a: 0, b: 10, c: 110, d: 111 B) a: 00, b: 01, c: 1, d: 111
 C) a: 1, b: 10, c: 00, d: 011 D) a: 10, b: 00, c: 01, d: 11

23. Which of the following is a possible Hamiltonian circuit of the below graph?

- A) {1, 6, 3, 4, 2, 5, 1} B) {1, 5, 2, 4, 3, 5, 1}
 C) {1, 5, 2, 4, 3, 6} D) {1, 5, 2, 4, 1}



24. The number of nodes in the state space tree to the m-coloring problem using backtracking is _____

- A) $m^{n+1} - 1$ B) $\frac{m^{n+1} - 1}{m - 1}$ C) $m - 1$ D) $2^{n+1} - 1$

25. Depth First Search is equivalent to which of the traversal in the Binary Trees?

- A) Pre-order B) Post-order C) Level-order D) In-order

26. If $f_1(n) \in O(g_1(n))$ and $f_2(n) \in O(g_2(n))$, then $f_1(n) + f_2(n) \in$

- A. $O(g_1(n))$ B. $O(g_2(n))$ C. $O(\max\{g_1(n), g_2(n)\})$ D. $\Theta(\max\{g_1(n), g_2(n)\})$

*** By using the dynamic programming approach to compute $\binom{6}{3}$ answer the following two questions

27. compute the value of binomial coefficient $\binom{6}{3}$ _____

- A) 25 B) 10 C) 20 D) 15

28. how many minimum terms that will be computed to find the value of $\binom{6}{3}$ _____

- A) 28 B) 27 C) 22 D) 21



***After creating pruned state space tree by using backtracking to solve the sum of subset problem with $n = 4$, $W = 15$, and $w = 3$ $w = 5$ $w = 6$ $w = 7$ answer the following two questions

19. The values of leaves in tree are

A) 0,5,11,3,9,8,15,14

B) 0,5,11,3,9,15

c) 14,15,8,9,6,12,5,0

d) 14,15,8,3,5,0

20. How many promising nodes in the tree

a) 7

b) 8

c) 10

d) 6

***** Write the correct answers of the Question # 1 in the table below:

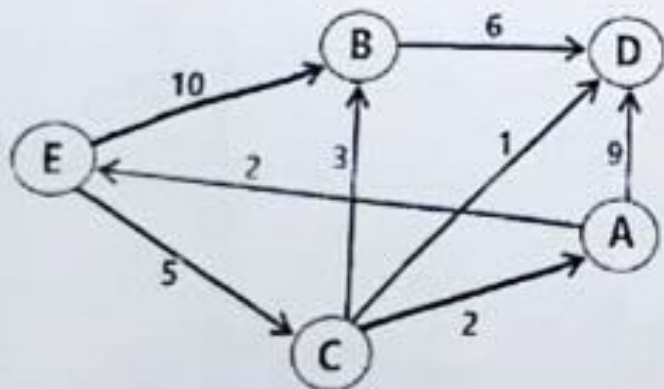
Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15
Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30

Question # 2: Based on the information of the below matrix, which represents the matrix generated by Floyd's algorithm at the 2-nd iteration (i.e. $D^{(2)}$) to find shortest paths between every pair of vertices in a specific weighted digraph, find the matrix which will be generated at the 3-rd iteration (i.e. $D^{(3)}$). [5 pts]

$$D^{(2)} = \begin{bmatrix} 0 & \infty & 6 & 3 & \infty \\ 3 & 0 & 9 & 6 & \infty \\ 4 & 1 & 0 & 2 & \infty \\ \infty & \infty & 1 & 0 & \infty \\ 7 & 13 & 4 & 2 & 0 \end{bmatrix}$$

$$D^{(3)} = \begin{bmatrix} 0 & 7 & 6 & 3 & \infty \\ 3 & 0 & 9 & 6 & \infty \\ 4 & 1 & 0 & 2 & \infty \\ 5 & 2 & 1 & 0 & \infty \\ 7 & 5 & 4 & 2 & 0 \end{bmatrix}$$

Question # 3: Based on the information given in the below graph, complete the below table after applying Dijkstra's Algorithm starting from vertex (A) [5 pts]



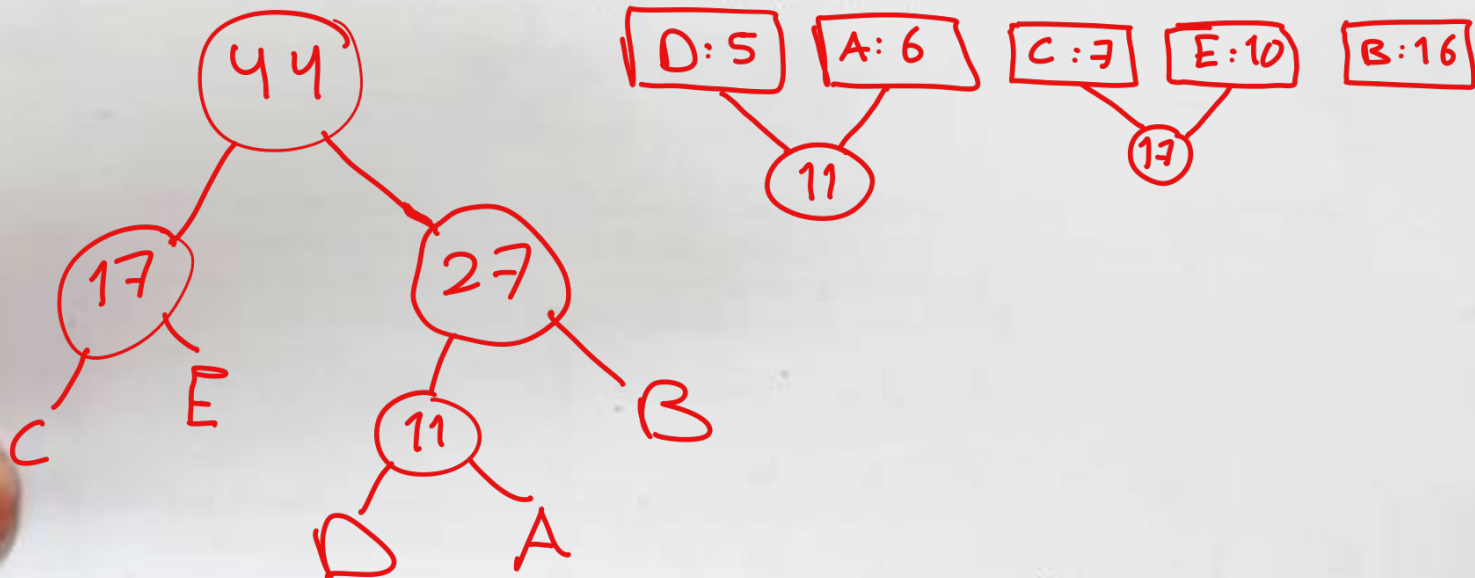
Selected vertex	B	C	D	E
A	∞	∞	∞	∞
E	∞	∞	9	1
C	12	7	9	2
D	10	7	8	2
B	10	7	8	2



Question # 4; Answer the following questions supposing a data file has characters and frequencies shown in the table below. If Huffman coding is used: [5 pts]

Character	Frequency
A	6
B	16
C	7
D	5
E	10

A- Construct the Huffman tree of the file (input) characters



B- Fill in the table below with the codeword for each character.

Character	Code
A	
B	
C	
D	
E	

A = 101

B = 11

C = 00

D = 100

E = 01

Solved By:

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